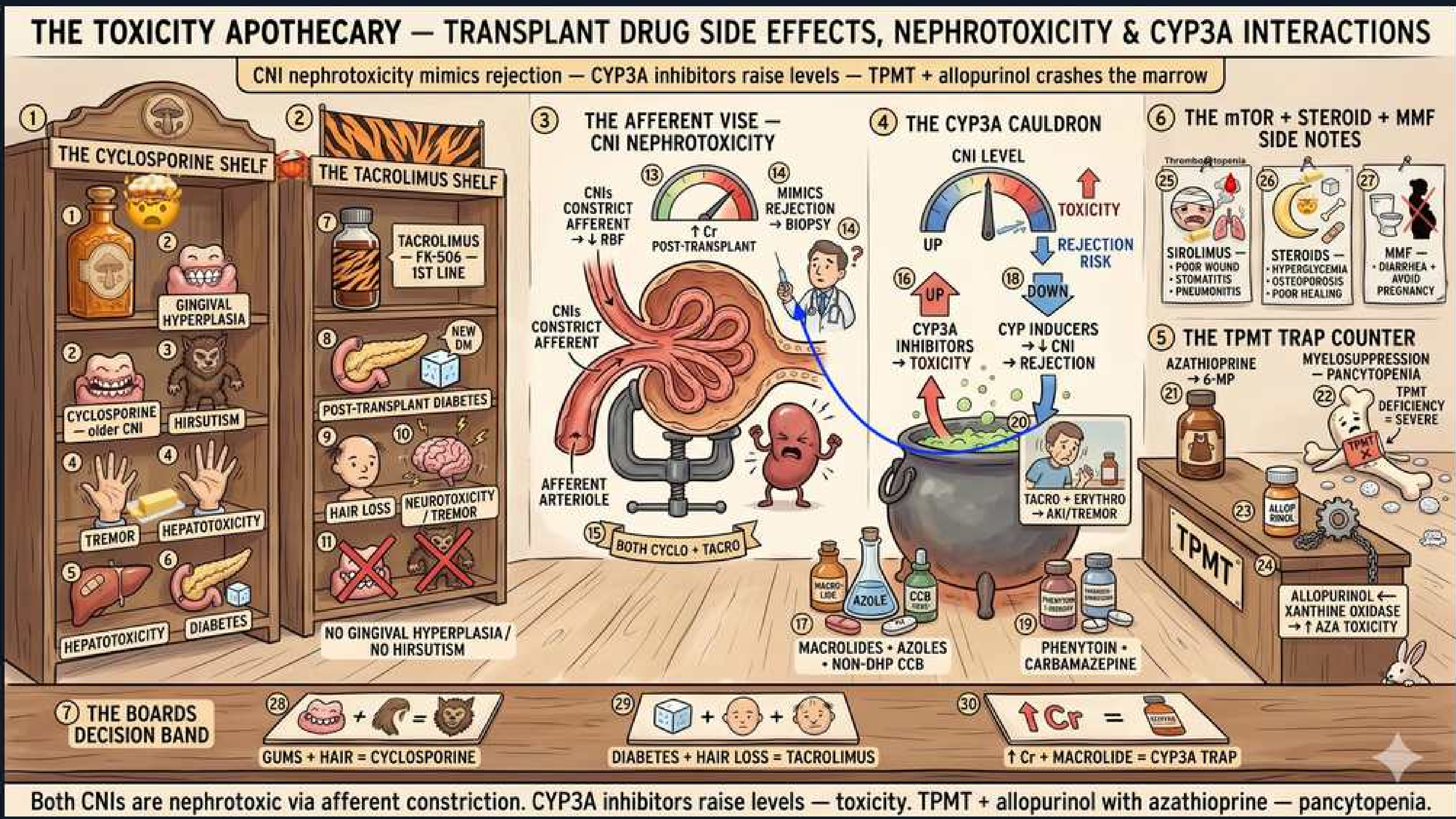
# Kidney Transplant — Drug Toxicities, Nephrotoxicity & CYP3A (Management 2)



## 1. Cyclosporine shelf

**WHAT YOU SEE**

The left apothecary shelf of bottles labelled CYCLOSPORINE — the OLDER, second-line calcineurin inhibitor

**WHAT IT ENCODES**

Cyclosporine is the older calcineurin inhibitor (CNI): it binds cyclophilin, and that complex inhibits calcineurin, blocking NFAT-driven IL-2 transcription and thereby T-cell activation. It is now generally second-line to tacrolimus because tacrolimus gives better graft survival and fewer rejection episodes. Both CNIs share a narrow therapeutic window and require trough-level therapeutic drug monitoring. Distinct cosmetic/metabolic side-effect signature vs tacrolimus is heavily tested.

**BOARD TRAP**

WRONG answer: calling cyclosporine first-line maintenance CNI. Tacrolimus is first-line; cyclosporine is reserved when tacrolimus is not tolerated. Discriminator: 'FK-506/tacrolimus = preferred first-line CNI; cyclosporine = older alternative.'

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## 2. Gingival hyperplasia

**WHAT YOU SEE**

Swollen overgrown gums drawn under the cyclosporine shelf — the classic cyclosporine cosmetic side effect (shared with phenytoin and dihydropyridine CCBs)

**WHAT IT ENCODES**

Gingival (gum) hyperplasia is a classic CYCLOSPORINE-specific adverse effect (the same triad of agents causing gum overgrowth: cyclosporine, phenytoin, and dihydropyridine CCBs like nifedipine/amlodipine). It is dose/level-related and improves with good oral hygiene and sometimes switching off cyclosporine. Tacrolimus does NOT cause gingival hyperplasia.

**BOARD TRAP**

WRONG answer: attributing gum overgrowth to tacrolimus. Gingival hyperplasia points to CYCLOSPORINE, not tacrolimus. Discriminator: a transplant patient with gum hypertrophy + hirsutism is on cyclosporine, not tacrolimus.

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## 3. Hirsutism

**WHAT YOU SEE**

A hairy bear-like figure under the cyclosporine shelf — excess hair GROWTH (hirsutism), the cyclosporine cosmetic clue (vs hair LOSS with tacrolimus)

**WHAT IT ENCODES**

Hirsutism (excess hair growth) is a CYCLOSPORINE-specific cosmetic effect, paired with gingival hyperplasia as the cyclosporine cosmetic duo. This is the opposite of tacrolimus, which tends to cause hair LOSS (alopecia). Useful exam discriminator since both drugs are CNIs with overlapping nephrotoxicity/neurotoxicity but divergent cosmetic profiles.

**BOARD TRAP**

WRONG answer: linking increased hair growth to tacrolimus. Hair GROWTH (hirsutism) = cyclosporine; hair LOSS (alopecia) = tacrolimus. Discriminator: direction of the hair change tells you which CNI.

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## 4. Tacrolimus shelf (FK-506)

**WHAT YOU SEE**

The second apothecary shelf labelled TACROLIMUS / FK-506 marked '1st LINE' — the preferred first-line calcineurin inhibitor

**WHAT IT ENCODES**

Tacrolimus (FK-506) is the first-line CNI: it binds FKBP-12, and that complex inhibits calcineurin → blocks NFAT → blocks IL-2 → suppresses T-cell activation (same downstream pathway as cyclosporine via a different binding protein). Preferred for superior graft survival and lower acute-rejection rates. Trough-level monitoring is mandatory (narrow window); both nephrotoxic and neurotoxic, and tacrolimus is the more diabetogenic of the two.

**BOARD TRAP**

WRONG answer: thinking tacrolimus and cyclosporine have entirely different mechanisms. They converge on calcineurin/NFAT/IL-2 — tacrolimus binds FKBP-12, cyclosporine binds cyclophilin. Discriminator: same target (calcineurin), different immunophilin.

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## 5. Post-transplant diabetes

**WHAT YOU SEE**

A glucose/diabetes icon under the tacrolimus shelf — tacrolimus is the more diabetogenic CNI, driving NODAT

**WHAT IT ENCODES**

Tacrolimus is more diabetogenic than cyclosporine and is a leading cause of NODAT (new-onset diabetes after transplant) by impairing pancreatic beta-cell insulin secretion; steroids add insulin resistance on top. NODAT raises cardiovascular and graft risk. Management: lifestyle, glycemic control, and consider lowering the tacrolimus level or switching to cyclosporine if diabetes is severe.

**BOARD TRAP**

WRONG answer: blaming new hyperglycemia post-transplant primarily on cyclosporine. NODAT favors TACROLIMUS over cyclosporine (steroids also contribute). Discriminator: post-transplant diabetes + tremor on a CNI → tacrolimus.

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## 6. Hair loss / tremor

**WHAT YOU SEE**

A balding head and a shaking/tremor figure under the tacrolimus shelf — alopecia plus fine tremor, tacrolimus's signature effects

**WHAT IT ENCODES**

Tacrolimus-specific effects: alopecia (hair LOSS, contrast with cyclosporine's hirsutism) and prominent neurotoxicity — fine tremor is the most common, plus headache, insomnia, and at high levels seizures or posterior reversible encephalopathy syndrome (PRES). Neurotoxicity is level-dependent, so a tremulous patient with a high trough should prompt dose reduction.

**BOARD TRAP**

WRONG answer: attributing a transplant patient's new tremor to cyclosporine while ignoring the tacrolimus level. Tremor/neurotoxicity is classically tacrolimus and is level-dependent. Discriminator: check the trough — high tacrolimus level + tremor → reduce dose.

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### 7. Both CNIs nephrotoxic

**WHAT YOU SEE**

An afferent arteriole squeezed in a 'vise' choking the kidney — both CNIs cause afferent vasoconstriction, and this nephrotoxicity MIMICS rejection (both raise Cr)

**WHAT IT ENCODES**

BOTH cyclosporine and tacrolimus are nephrotoxic via afferent arteriolar vasoconstriction, reducing renal blood flow and GFR (acute, dose/level-dependent and reversible) and, chronically, causing interstitial fibrosis/tubular atrophy with arteriolar hyalinosis. Critically, CNI nephrotoxicity MIMICS acute rejection — both present as a rising creatinine in a transplant patient — so the two must be distinguished by trough level and allograft biopsy before acting.

**BOARD TRAP**

WRONG answer: reflexively treating every creatinine rise on a CNI as acute rejection and pulsing steroids. A high CNI trough suggests drug nephrotoxicity (treat by lowering dose), whereas a low trough/positive biopsy suggests rejection. Discriminator: rising Cr on a CNI → check trough level AND biopsy before deciding rejection vs toxicity.

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### 8. CNI level → rejection vs toxicity

**WHAT YOU SEE**

A bubbling cauldron of CNI 'level': filled too high = TOXICITY, drained too low = REJECTION risk — the narrow therapeutic window read off the trough

**WHAT IT ENCODES**

CNIs have a narrow therapeutic window, so trough therapeutic drug monitoring is mandatory: a HIGH level → nephrotoxicity/neurotoxicity; a LOW level → under-immunosuppression and rejection risk. Because both extremes raise creatinine, the trough plus biopsy is what separates 'too much drug' (toxicity) from 'too little drug' (rejection). Any drug or food (grapefruit) that shifts CNI metabolism can swing the patient into either zone.

**BOARD TRAP**

WRONG answer: ignoring the trough level when creatinine changes. The trough is the pivotal data point — high = toxicity, low = rejection. Discriminator: never decide rejection vs toxicity without the CNI level.

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### 9. CYP3A inhibitors ↑ level

**WHAT YOU SEE**

Azole and macrolide bottles tipping INTO the cauldron, raising the level — CYP3A4 inhibitors push CNI troughs into the toxic range

**WHAT IT ENCODES**

CNIs are metabolized by CYP3A4 (and transported by P-glycoprotein), so CYP3A4 INHIBITORS RAISE CNI levels into the toxic range. High-yield inhibitors: azole antifungals (fluconazole, voriconazole, itraconazole, ketoconazole), macrolides (erythromycin, clarithromycin — NOT azithromycin), non-dihydropyridine CCBs (diltiazem, verapamil), protease inhibitors, and grapefruit juice. When you add any of these, anticipate a higher trough → reduce the CNI dose and recheck levels.

**BOARD TRAP**

WRONG answer: starting fluconazole or clarithromycin in a transplant patient without adjusting the CNI. These CYP3A inhibitors spike tacrolimus/cyclosporine into nephrotoxic/neurotoxic range. Discriminator: azole/macrolide/diltiazem added → preemptively lower CNI dose and monitor trough.

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### 10. CYP3A inducers ↓ level

**WHAT YOU SEE**

Phenytoin / rifampin / carbamazepine draining the cauldron level DOWN — CYP3A4 inducers drop CNI troughs toward rejection

**WHAT IT ENCODES**

CYP3A4 INDUCERS LOWER CNI levels, risking sub-therapeutic immunosuppression and acute rejection. High-yield inducers: rifampin (the classic, most potent), the anticonvulsants phenytoin, carbamazepine, and phenobarbital, and St. John's wort. Starting one of these can drop the tacrolimus trough days later — increase the CNI dose and monitor, or choose a non-inducing alternative (e.g. treat latent TB without rifampin where feasible).

**BOARD TRAP**

WRONG answer: adding rifampin (e.g. for TB) to a transplant regimen and keeping the CNI dose unchanged. Rifampin induction can crash the trough and precipitate rejection. Discriminator: rifampin/phenytoin/carbamazepine added → CNI level falls → rejection risk → raise dose or avoid the inducer.

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### 11. mTOR / sirolimus side effects

**WHAT YOU SEE**

A sirolimus side-note panel showing poor wound healing, proteinuria, mouth ulcers and pneumonitis — the mTOR-inhibitor toxicity signature

**WHAT IT ENCODES**

mTOR inhibitors (sirolimus, everolimus) block the mTOR pathway downstream of IL-2, halting T-cell proliferation. Signature toxicities: impaired/delayed WOUND HEALING and lymphocele, proteinuria, oral (mouth) ulcers/stomatitis, hyperlipidemia, pneumonitis (interstitial lung disease), and cytopenias. They are non-nephrotoxic (no afferent vasoconstriction), so they're sometimes used to spare CNI in chronic toxicity, and sirolimus has an antineoplastic effect useful when post-transplant skin cancer recurs.

**BOARD TRAP**

WRONG answer: starting sirolimus in the immediate post-operative period. Its impaired wound healing causes wound dehiscence and lymphocele — avoid early post-op. Discriminator: defer mTOR inhibitors until the surgical wound has healed; new proteinuria or pneumonitis on a transplant patient points to an mTOR inhibitor.

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### 12. Steroid / MMF side effects

**WHAT YOU SEE**

A steroid + mycophenolate side-note panel showing diabetes, stomatitis and GI upset — the steroid and MMF adverse profiles

**WHAT IT ENCODES**

Glucocorticoids contribute to hyperglycemia/NODAT, osteoporosis and avascular necrosis, weight gain, cataracts, hypertension, and infection risk. Mycophenolate (MMF/mycophenolic acid), the workhorse antiproliferative that blocks IMP dehydrogenase (purine synthesis in lymphocytes), mainly causes dose-related GI toxicity (diarrhea, nausea, abdominal pain) and myelosuppression (leukopenia/anemia); it is also teratogenic. MMF is the first agent reduced when treating infection or marrow suppression.

**BOARD TRAP**

WRONG answer: attributing severe diarrhea on the transplant regimen to an infection only, without considering mycophenolate. MMF is a leading drug cause of post-transplant diarrhea. Discriminator: dose-related GI symptoms/leukopenia on the regimen → think mycophenolate (and consider dose reduction or CMV colitis on the differential).

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## 13. TPMT trap counter

**WHAT YOU SEE**

An azathioprine→6-MP counter with a 'myelosuppression / pancytopenia' warning — low TPMT can't clear the active thiopurine, crashing the marrow

**WHAT IT ENCODES**

Azathioprine is a prodrug → 6-mercaptopurine; its inactivation depends on the enzyme TPMT (thiopurine methyltransferase). Patients with low/absent TPMT activity cannot clear the active thiopurine metabolites and develop severe, sometimes fatal myelosuppression/pancytopenia. Therefore TPMT activity (and/or genotype) should be checked before starting azathioprine, with dose reduction or avoidance in deficiency, plus routine CBC monitoring.

**BOARD TRAP**

WRONG answer: starting standard-dose azathioprine without assessing TPMT and then blaming the resulting pancytopenia on infection. Low TPMT → thiopurine accumulation → marrow failure. Discriminator: check TPMT before azathioprine; profound cytopenia after azathioprine = TPMT deficiency until proven otherwise.

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## 14. Allopurinol + azathioprine

**WHAT YOU SEE**

Allopurinol blocking the xanthine-oxidase pathway, shunting azathioprine to toxic accumulation — the classic interaction causing pancytopenia

**WHAT IT ENCODES**

Azathioprine/6-MP is also catabolized by xanthine oxidase. Allopurinol (and febuxostat) inhibit xanthine oxidase, so co-administration causes 6-MP/thioguanine nucleotides to accumulate → severe myelosuppression and pancytopenia. This is a classic board interaction. If both are truly needed, azathioprine must be reduced to roughly 25–33% of the usual dose (or, better, avoid the combination / use an alternative urate-lowering approach).

**BOARD TRAP**

WRONG answer: adding allopurinol for gout to a patient on full-dose azathioprine without dose adjustment. Xanthine oxidase inhibition triggers thiopurine toxicity and marrow suppression. Discriminator: allopurinol + azathioprine → reduce azathioprine to ~25% or avoid; new pancytopenia after starting allopurinol on azathioprine is this interaction.

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